Iman Rahmaty

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Research Interests: Distributed Systems, Wireless Communication, IoT, Mobile Edge Computing (MEC), Performance Optimization, Multi-Agent Reinforcement Learning (RL), Federated/Collaborative Learning

EDUCATION

Master of Science in Computer Engineering/Computer Networks Graduated Sep. 2022

GPA: 87/100 Sharif University of Technology (SUT), Tehran, Nationally ranked no. 1 in Iran

Thesis: Distributed Computation Offloading based on Deep RL (DRL) in Resource-Constrained MEC

Supervisor: Prof. Ali Movaghar

Bachelor of Science in Industrial Engineering Graduated Sep. 2019

Khajeh Nasir Toosi University of Technology (KNTU), Tehran, Nationally ranked no. 5 in Iran

RESEARCH EXPERIENCE

Research Engineer at EdgeAI Laboratory Tehran, Iran, 2022 - Present

Supervisor: Prof. Hamed Shah-Mansouri Department of Electrical Engineering, SUT

Research Theme: Multi-agent DRL for collaborative computation offloading in partial observability MEC systems with a focus on scalable and joint decision-making among heterogeneous agents under non-stationary system dynamics. Targeting to enhance resource utilization and load balancing through collaborative offloading scenarios, while enabling agents to learn cooperative strategies through centralized training with decentralized execution (CTDE).

Research Assistant at Performance and Dependability Laboratory (PDL) 2019-2022

Supervisor: Prof. Ali Movaghar Department of Computer Engineering, SUT

Research Theme: DRL-based task offloading in highly-dynamic and energy-constrained MEC systems, focusing on distributed and independent offloading decision-making under time-varying network conditions for time-sensitive tasks with strict processing deadlines. Aiming to maximize end-user QoE by prioritizing and balancing the trade-offs between task completion, latency, and energy consumption according to individual requirements, while accounting for dynamic workloads at edge nodes.

PUBLICATION

- **I. Rahmaty**, H. Shah-Mansouri, A. Movaghar, "QECO: A QoE-Oriented Computation Offloading Algorithm based on Deep Reinforcement Learning for Mobile Edge Computing", *IEEE Transactions on Network Science and Engineering*, vol. 12, no. 4, pp. 3118-3130, July-Aug. 2025.
- **I. Rahmaty**, H. Shah-Mansouri, M. Hosseini, A. Movaghar, "A Collaborative Computation Offloading Framework based on Multi-Agent Deep Reinforcement Learning in Partially Observable Mobile Edge Computing Environments", *Work in progress.*

Throughout these projects, I was fully involved in all stages, including problem formulation, system and methodology design, algorithm development, implementation, evaluation, and manuscript writing, which strengthened my understanding of dynamic system optimization via advanced DRL approaches.

HONORS

❖ Ranked top 10% of M.Sc. students in the Dept. of Computer Engineering at SUT (Class of 2019) 2022
❖ Ranked 55th among 60,000 in the nationwide M.Sc. entrance exam in Computer Engineering. 2019
❖ Ranked top 1% among 180,000 in the nationwide B.Sc. entrance exam in Mathematics and Physics. 2014
❖ Achieving the 3rd position in the RoboCup Competition (IranOpen) 2012

SELECTED COURSES

- Computer Performance Evaluation	$\mathcal{A}^+$	- Theory of Distributed Systems	$\mathcal{A}$
- Verification of Reactive Systems	$\mathcal{A}^+$	- Computer Network	$\mathcal{A}$
- Advanced Network Security	$\mathcal{A}^+$	- IT Enterprise architecture	$\mathcal{A}^+$
- Wireless Networking	$\mathcal{A}^+$	- Computer Network Management	$\mathcal{B}^+$

ACADEMIC EXPERIENCE

Reviewer, IEEE Internet of Things Journal IEEE, 2025

Reviewer, 27th International Computer Conference CSICC, 2022
Computer Society of Iran (CSICC)

Teaching Assistants

- **Performance Evaluation of Computer Systems** (Head TA) SUT, 2020-2022
  Prof. Ali Movaghar and Dr. Mahdi Dolati
- **Software Defined Networking** (Head TA) SUT, 2022
  Prof. Ali Movaghar and Dr. Mohammad Hosseini
- **Verification of Reactive Systems** SUT, 2021
  Prof. Ali Movaghar
- **Theory of Machines and Languages** (Head TA) SUT, 2021
  Prof. Ali Movaghar
- **Wireless Networking** SUT, 2020
  Prof. Ali Mohammad Afshin Hemmatyar

SKILLS

- **Professional:** Stochastic network modeling and simulation, performance evaluation, wireless and distributed network optimization, RL-driven decision-making under uncertainty and limited information
- **Programming Languages:** Python, R, Bash, C++, SQL
- **Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-learn
- **Data Analysis:** Pandas, NumPy, Matplotlib, Jupyter Notebooks
- **Frameworks & Tools:** Linux, Mininet, Ns-3, Git, LaTeX, Vim, Flask, Visio
- **Language Proficiency:** Persian (Native), English (Working proficiency)
  - TOEFL (IBT), Score: 108/120 (R: 30, L: 28, S: 22, W: 28), 2 Jan 2024

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SELECTED PROJECTS

- **Multi-Agent Actor–Critic Network Based on CTDE** EdgeAI, 2025
  Developed a hierarchical multi-agent DRL algorithm for collaborative decision-making in non-stationary environments, based on decentralized partially observable markov decision processes (Dec-POMDPs).
- **Collaborative Cloud-Edge-End MEC Environment** EdgeAI, 2024
  Decision-driven simulation of communication-assisted MEC systems, evaluating collaborative computation offloading across cloud servers, edge nodes, and end devices.
- **Dueling Double Deep Q-Network (D3QN)** EdgeAI, 2023
  Developed a D3QN algorithm for distributed decision-making under uncertainty, leveraging both double Q-learning and dueling network architectures. Designed for personalized QoE maximization in dynamic MEC environments based on markov decision processes (MDPs).
- **Long Short-Term Memory (LSTM)** PDL, 2022
  Designed an LSTM model to continuously estimate dynamic workloads at edge servers, dealing with limited global information and uncertainty in Q-learning-based decision-making in MEC.
- **Resource-constrained MEC Environment** PDL, 2021
  Time-driven simulation of computation and communication processes in queuing-based MEC systems, modeling stochastic system conditions and enabling evaluation of task offloading strategies.
- **Queueing System** SUT, 2020
  Event-driven simulation and performance evaluation of M/M/1/K queues under FCFS, processor sharing (PS), and discriminatory PS service disciplines.

CERTIFICATION

Interactive Learning Tehran Institute for Advanced Studies (TeIAS), 2021
Certification of completion in DRL course, Instructor: Prof. Majid Nili
Machine Learning and Deep Learning in Python Start-Tech Academy, 2020
Certification of completion in Udemy online course
Data Science Tose'e Higher Education Institute, 2019
Certification of completion in Data Science course, Instructor: Dr. Yaser Zerehsaz
Advanced Python Topics Remis Arjang Institute, 2018
Certification of completion in Advanced Python course, Instructor: Dr. Peyman Hooshmandi
LPIC1 Anisa Iran Linux House, 2017
Certification of completion in Linux Administrator course, Instructor: Dr. Amir Abbasi

REFERENCES

Prof. Ali Movaghar movaghar@sharif.edu
Professor of Computer Engineering, SUT, Tehran, Iran.
Prof. Hamed Shah-Mansouri hamedsh@sharif.edu
Assistant Professor of Electrical Engineering, SUT, Tehran, Iran.
Prof. Mohammad Hosseini m-hosseini@sbu.ac.ir
Assistant Professor of Computer Engineering, Shahid Beheshti University, Tehran, Iran.
Prof. Ali Mohammad Afshin Hemmatyar hemmatyar@sharif.edu
Professor of Computer Engineering, SUT, Tehran, Iran.